SERGEI CHUPROV

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Google Scholar **LinkedIn**

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PhD in Computing and Information Sciences

The University of Texas Rio Grande Valley (UTRGV), Edinburg, TX

RESEARCH AREAS OF INTEREST

Artificial Intelligence and Machine Learning Systems, Robustness of Intelligent Systems and their Practical Applications, Cybersecurity, Data Quality and Security Evaluation and Analytics

HIGHLIGHTS

- 53 peer-reviewed publications, 2 pending patent applications, Google Scholar ***h***-index: 9.
- My publication list includes renowned peer-reviewed conferences and journals, such as IEEE TDSC journal, IEEE ICC, IEEE Systems journal, IEEE CEC, IEEE WCCI, IEEE INFOCOM (CNERT Workshop), and others.
- I have a valuable experience participating in R&D projects supported by NSF, DoD (USMA), CRDF Global, and other funding agencies.
- I received multiple grants and scholarships awarded for the excellence demonstrated in research and education.
- I have 7+ years of Graduate Research and Teaching Assistant experience, mentoring and advising students in various institutions and countries.

EDUCATION

PhD in Computing and Information Sciences
Rochester Institute of Technology (RIT), NY, USA *August 2021 - May 2024*
Department of Computer Science and Department of Cybersecurity **GPA:** 3.83/4.0
Thesis: Robust Machine Learning Under Vulnerable Cyberinfrastructure and Varying Data Quality

Master's in Information Security
Saint-Petersburg National Research University of Information Technologies, Mechanics and Optics (ITMO), Russia *September 2017 - June 2019*
GPA: 4.0/4.0 (with honors)
Faculty of Secure Information Technologies
Thesis: Assurance of Information in a Mobile Robotic System with Decentralized Control

Bachelor's in Information Security
ITMO University, Saint Petersburg, Russia *September 2013 - June 2017*
Faculty of Secure Information Technologies **GPA:** 3.5/4.0
Thesis: Vulnerability Analysis and Classification in Local Networks Based on Microsoft OS Family

ACADEMIC WORK & TEACHING EXPERIENCE

Department of Computer Science, UTRGV
Edinburg, TX, USA *September 2024 - present*
Assistant Professor (tenure-track)
Scope & Responsibilities: teach undergraduate and graduate courses, conduct research in AI/ML security and robustness, and contribute to departmental and university service activities. **Courses (full-time instructor):**

- CSCI 2380 Computer Science II (undergraduate)
- CSCI 4341 Foundations of Intelligent Security Systems (undergraduate)
- CSCI 6370 Foundations of Intelligent Security Systems (graduate)

B. Thomas Golisano College of Computer Science, RIT
Rochester, NY, USA *August 2021 - July 2024*
Graduate Research Assistant (part-time)
Scope & Responsibilities: research and development in the area of Artificial Intelligence- and Machine Learning-based systems security, Data Quality and Security Assurance, Robustness of Machine Learning-end Systems. **Results:** 2 pending patent applications; more than 15 published papers in refereed journals and conference proceedings (see Publication List below).

B. Thomas Golisano College of Computer Science, RIT
Rochester, NY, USA *August 2022 - July 2024*
Graduate Teaching Assistant (part-time)
Scope & Responsibilities: assisting the course instructor in teaching Computer Science-related courses. **Results:** gained teaching activities experience and knowledge related to building effective communication with the students; holding office hours and grading assignments for the classes of 20 undergraduate and graduate students; providing constructive feedback and communicating students' questions and concerns related to the course content and assignments; supervised student capstone projects and mentored them. **Courses:**

- CSCI-735 Foundations of Intelligent Security Systems (instructor: Dr. Leon Reznik)
- CSCI-536 Information Retrieval, undergraduate section (instructor: Dr. Richard Zanibbi)
- CSCI-636 Information Retrieval, graduate section (instructor: Dr. Richard Zanibbi)

Mobile Intelligent Systems Laboratory, ETU "LETI" University
Saint Petersburg, Russia *May 2021 - August 2021*
Researcher (part-time)
Scope & Responsibilities: research and development in the area of Cyber-Physical Systems security and safety, Intelligent Systems security. Lead a group of 4 developers and engineers; developed models and software tools for Reputation and Trust-based simulation environment to model communication in decentralized IoT networks; applied the developed models and tools on a real-world IoT hardware devices; developed projects' reports and technical documentation. **Results:** 7 published papers in refereed journals and conference proceedings (see Publication List below).

Department of Secure Information Technologies, ITMO University
Saint Petersburg, Russia *September 2019 - May 2021*
Teaching Assistant (part-time)
Scope & Responsibilities: assisting the course instructor in teaching undergraduate and graduate courses in the areas of Information Security and Computer Science. **Results:** gained experience in teaching laboratory and practical sessions for the classes of 10-20 undergraduate and graduate students; holding office hours and communicating students' questions and concerns; developed and graded assignments for the selected courses; assisted in developing the coursework content for the "Functional Security and Safety Assurance in Autonomous Vehicles" Master's degree program, established at ITMO in 2019. **Courses:**

- Foundations of Information Technology (instructor: Dr. Ilia Viksnin)
- Introduction to Imitation Modeling (instructor: Dr. Igor Komarov)
- Information Security Technology and Methodology (instructor: Dr. Victoria Korzhuk)
- Foundations of Autonomous Vehicles Safety (instructor: Dr. Ilia Viksnin)
- Autonomous Vehicles Imitation Modeling (instructor: Dr. Cyril Pshenichny)

Department of Secure Information Technologies, ITMO University
Saint Petersburg, Russia *May 2018 - May 2021*
Researcher (part-time)
Scope & Responsibilities: research and development in the area of Information Security, Multi-Agent Mobile Robotic Systems Security, Knowledge Representation in Intelligent Systems. **Results:** developed security and safety models and methods for decentralized and multi-agent mobile robotic systems (see Projects below); developed security-inspired smart factory models in AnyLogic simulation environment; 14 publications in refereed journals and conference proceedings (see Publication List below).

RESEARCH & DEVELOPMENT PROJECTS

Collaborative research: IDEAS lab: ETAUS: Smarter Microbial Observatories for Realtime ExperimentS (SMORES) (non PI or Co-PI)

August 2023 - Present

- **Objective:** this recently started project involves the collaboration between RIT, Harvard University, Florida International University, and the University of Georgia to study marine sediments and their role in natural carbon sequestration, using AI and ML to optimize data collection and predictions during field experiments. The project is supported by NSF award #2321652.

Enhancing Security and Privacy of the Conventional Federated Learning (non PI or Co-PI)

September 2022 - Present

- **Objective:** to develop methods and software tools aimed at detecting compromised and failed local units in the Federated Learning and excluding them from the aggregation. The project is supported by the United States Military Academy (USMA) and is accomplished under Grant Number W911NF-20-1-0337.

Developing Methods and Tools for Machine Learning Robustness Assurance under Vulnerable Cyberinfrastructure and Varying Data Quality (non PI or Co-PI) *May 2022 - April 2024*

- **Objective:** to develop system integration methods and tools for providing feedback to the Machine Learning system cyberinfrastructure components in order to enhance Machine Learning robustness to input Data Quality variations. The project was supported by the grant from the U.S. Civilian Research & Development Foundation (CRDF Global) with funding from the United States Department of State.

Partisan Telegram Application and Operational Security Assessment (non PI or Co-PI)

February 2022 - July 2022

- **Objective:** to conduct comprehensive security evaluation of the Partisan Telegram Android application. **Results:** developed an open-source report on comprehensive security assessment of the Partisan Telegram Android OS application; conducted functional analysis of the application's source code; investigated and verified vulnerabilities that jeopardized user's privacy. The project was supported by the Open Technology Fund.

Enhancing Sensor Network Security with Reputation and Trust-Based Technique (non PI or Co-PI)

May 2021 - August 2021

- **Objective:** to develop methods and software tools aimed at detecting malicious nodes in sensor networks using Reputation and Trust indicators. The project was supported by the Ministry of Science and Higher Education of the Russian Federation, #075-01024-21-02 dated 29.09.2021 (project FSEE-2021-0014)

Development of Hardware and Software Modules for Critical Objects Monitoring over the Conditions of Uncertainty and Data Incompleteness for Malicious Attacks Prevention and Mitigation, *Industrial partnership project* (non PI or Co-PI) *September 2019 - October 2020*

- **Objective:** to develop software methods and tools implemented on sensor-equipped hardware modules aimed at monitoring and detecting hazardous environmental events in the conditions of malicious attacks. **Results:** developed intelligent security evaluation models, risk evaluation models and methods, and software and hardware prototypes, implemented by Murmansk Commercial Seaport OAO, Russia. The work was supported by the Ministry of Science and Higher Education of the Russian Federation under the agreement No. 05.605.21.0189.

Development of a Quantum Secure Hybrid Platform for Distributed Cyber-Physical Systems Control (non PI or Co-PI) *May 2018 - August 2019*

- **Objective:** to develop methods and tools aimed at assuring data security and confidentiality in a group of autonomous mobile robotic devices equipped with quantum key generation facilities. **Results:** developed Multi-Agent based security models and methods for mobile robotic system equipped with quantum key generation facilities, and mobile robotic devices prototypes, implemented at ITMO University for further research and educational purposes. The work was financially supported by the Government of Russian Federation, Grant 074-U01.

Development of Experimental Testbed for Smart City Control and Security Algorithms Research and Verification (non PI or Co-PI) *May 2018 - March 2019*

- **Objective:** to develop physical testbed for conducting real-world empirical verification of the developed IoT security solutions deployed in decentralized systems, such as Intelligent Transportation Systems. The video of the developed testbed can be found in a public access: URL.

GRANTS AND AWARDS

2025 Teach Access Fellowship *December 2024*
USA

IEEE CAI 2023 Student Conference Grant *June 2023*
Santa Clara, CA, USA

Gold Medal (1st prize) for the poster presented at the UPSTAT 2023 conference *April 2023*
Rochester, NY, USA

NSF grant for participation in POWDER Mobile and Wireless Week 2023 *January 2023*
Salt Lake City, UT, USA

Amazon AWS Cloud Credit for Research, awarded for one year *August 2022*
Rochester, NY, USA

Saint-Petersburg's Academic and Industrial Institutions Students' Grant *(2): 2019, 2022*
Saint-Petersburg, Russia

IEEE INFOCOM 2022 Student Conference Grant *May 2022*
London, UK

Rochester Institute of Technology PhD Research and Tuition Scholarship *(3): 2021-2023*
Rochester, NY, USA

Scholarship of the Russian Federation President for Students of Higher Education Universities Studying Abroad in 2021/2022 Academic Year *May 2021*
Saint-Petersburg, Russia

Scholarship of the Russian Federation President and Government for the Students of Priority Education Directions *(2): 2021, 2022*
Saint-Petersburg, Russia

ITMO University's Best Student's Research MS Thesis Award *August 2019*
Saint-Petersburg, Russia

Scholarship of the Russian Federation Government *September 2019*
Saint-Petersburg, Russia

INVITED TALKS & SESSIONS

Invited Talk (remote) at From Theory to Practice (T2P) Worshop, Kigali, Rwanda *16 April 2024*
Title: *Security and Robustness of Machine Learning under Vulnerable Cyberinfrastructure and Varying Data Quality*

Invited Session at UPSTAT 2024 Conference, NY, USA *13 April 2024*
Title: *Improving Federated Learning Robustness towards Security Violations and Data Quality Degradations OR How to Learn Better via Extracting Knowledge and Accumulating History?*
Organized together with Dr. Leon Reznik and Raman Zatsarenko (PhD student, RIT)

SERVICE, VOLUNTEERING & MENTORING EXPERIENCE

◇ ACADEMIC SERVICE:

IEEE Transactions on Artificial Intelligence
Reviewer: 2025

2025 International Joint Conference on Neural Networks (IJCNN 2025)
Reviewer: 2025

IEEE IoT Journal
Reviewer: 2024, 2025

MDPI Sensors
Reviewer: 2024, 2025

MDPI Network
Reviewer: 2024, 2025

MDPI Electronics
Reviewer: 2024, 2025

Journal of Data and Information Quality
Reviewer: 2024, 2025

IEEE World Congress on Computational Intelligence (WCCI) 2024: The International Joint Conference on Neural Networks (IJCNN)
Reviewer: 2024

The 30th International Conference on Neural Information Processing (ICONIP 2023)
Reviewer: 2023

◇ VOLUNTEERING:

Engaged Scholar Symposium
The University of Texas Rio Grande Valley, TX, USA *2025*
Abstract Reviewer and Judge
Evaluated and provided feedback for the abstracts and projects presented by students.

GENIUS Olympiad
Rochester Institute of Technology, NY, USA *(2): 2023, 2024*
Volunteer Judge
Evaluated and graded 12 science projects (each year) presented at 2023 and 2024 GENIUS Olympiad. Provided constructive feedback on the projects and presentations to the participants.

◇ STUDENT MENTORING:

Angel Peredo, PhD
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2025*

Danny McClain, BS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2025*

Ei Ei Nyein Chan, MS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2025*

Damian Villarreal, BS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2025*

Bryn Ramirez, MS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2025*

Nathaniel Deleon, BS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2025*

Kayla Luevanos, BS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2025*

Andrew Alwarez, BS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2025*

Gael Marquez, BS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2025*

Adrian Pena, MS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2024, 2025*

Pedro Fonseca, PhD
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2024, 2025*

Eduardo Dominguez, BS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2024, 2025*
Supported by UTRGV Engaged Scholar Award for Undergraduate Research (with Dr. Chuprov as a mentor)

Carlos Sepulveda, BS
The University of Texas Rio Grande Valley, Edinburg, TX, USA *2024*

Harshil Pravinbhai Patel, MS
Rochester Institute of Technology, NY, USA *2024*
MS capstone project: "Improving Federated Learning Security with Trust Evaluation to Detect Adversarial Attacks"

Shweta Sandip Sharma, MS
Rochester Institute of Technology, NY, USA *2024*
MS capstone project: "Analyzing Effects of Adversarial Attacks on Classifier Performance"

Gaurav Thakur, MS
Rochester Institute of Technology, NY, USA *2024*
MS capstone project: "Resilience of Federated Learning Models in Image Classification Under Data Degradation"

Vedika Vishwanath Painjane, MS
Rochester Institute of Technology, NY, USA *2024*
MS capstone project: "Analyzing the Effect of Data Poisoning on Semantic Segmentation Using Federated Learning"

Anirudh Narayanan, MS
Rochester Institute of Technology, NY, USA *2024*
MS capstone project: "Soccer Games Automatic Video Highlights Generator", [GitHub]

Moinuddin Memon, MS
Rochester Institute of Technology, NY, USA *2023*
MS capstone project: "Knowledge-Based Client Filtering Algorithm for Federated Learning" (see [C10])

Ilya Sakhno, BS
ITMO University, Saint Petersburg, Russia *2021*
BS qualification project: "Unified Security Profile for Contactless Payment Technology"

Egor Marinenkov, BS
ITMO University, Saint Petersburg, Russia *2020*
BS qualification project: "Detecting Data Integrity Violations in Cyber-Physical Systems using Game Theory Approach" (see [J5] and [C31])

ADDITIONAL RESEARCH EXPERIENCE

Summer research program in Cybersecurity under the supervision of Dr. Leon Reznik
RIT, NY, USA *June 2019 - August 2019*
Results: 3 published papers [J6,C19,C20] related to security and safety assurance in a group of autonomous vehicles with Reputation, Trust, and Data Quality indicators.

PUBLICATION LIST

PATENTS

[P1] **Chuprov, S.**, & Reznik, L. "Federated Learning with A Compromised Unit Exclusion from Receiving Global Model Updates".

Provisional application filed on January 13, 2023. Converted to non-provisional

[P2] **Chuprov, S.**, & Reznik, L. "Network Adjustment based on Machine Learning End System Performance Monitoring Feedback".

Provisional application filed on September 14, 2022. Converted to non-provisional

BOOKS & BOOK CHAPTERS

[B1] **Chuprov, S.**, Zatsarenko, R., & Reznik, L. (2025) "mLINK: Machine Learning Integration with Network and Knowledge" in Metacognitive Artificial Intelligence, eds. Shakarian, P. & Wei, H. Cambridge University Press. *To be published in September 2025.* URL

REPRINTS

[R1] **Chuprov, S.**, Belyaev, P., Gataullin, R., Reznik, L., Neverov, E., & Viksnin, I. (2024) "Robust Autonomous Vehicle Computer-Vision-Based Localization in Challenging Environmental Conditions" in Applied Sciences. 2024, pp. 34-48., doi: doi.org/10.3390/books978-3-7258-2184-6. *Reprint of the Special Issue "Challenges in the Guidance, Navigation and Control of Autonomous and Transport Vehicles" that was published in Applied Sciences in 2023.* URL

REFEREED JOURNAL PUBLICATIONS

[J1] Korobeinikov, D., Zatsarenko, R., **Chuprov, S.**, Barea, A., & Reznik, L. (2026). "InteFL Framework: Optimizing Federated Learning with Metacognition for Application Design and Deployment" in IEEE Intelligent Systems. 2026. doi: 10.1109/MIS.2026.3658072. URL. **Q1**. *Published in Special Issue*

[J2] **Chuprov, S.**, Reznik, L., & Zatsarenko, R. (2025). "KISS: Knowledge Integration System Service for ML End Attacks Detection and Classification" in IEEE Transactions on Dependable and Secure Computing. 2025, pp. 1-12. doi: 10.1109/TDSC.2025.3553807. URL. **Q1**.

[J3] **Chuprov, S.**, Zatsarenko, R., Reznik, L., & Khokhlov, I. (2024). "Data Quality Based Intelligent Instrument Selection with Security Integration" in ACM Journal of Data and Information Quality. 2024, vol. 16, no. 3, pp. 1-24. doi: 10.1145/3695770. URL. **Q2**.

[J4] **Chuprov, S.**, Belyaev, P., Gataullin, R., Reznik, L., Neverov, E., & Viksnin, I. (2023). "Robust Autonomous Vehicle Computer-Vision-Based Localization in Challenging Environmental Conditions" in Applied Sciences. 2023, no. 13(9):5735. doi: 10.3390/app13095735. URL. **Q1**. *Published in Special Issue*

[J5] Berezovskaya, O., **Chuprov, S.**, Neverov, E., & Sadreev, E. (2023). "Review and Comparison of Lightweight Modifications of the AES Cipher for a Network of Low-Power Devices" in Automatic Control and Computer Sciences. 2022, no. 56, pp. 994-1006. doi: 10.3103/S0146411622080028. URL

[J6] Melnikov, T., **Chuprov, S.**, Lazarev, E., Gataullin, R., & Viksnin, I. (2022). "Improving Reputation and Trust-Based Approach with Reliability Indicators for Autonomous Vehicles Intergroup Communication" in Tomsk State University Journal of Control and Computer Science. 2022, no. 61, pp. 108-117. URL

[J7] Marinenkov, E., **Chuprov, S.**, Tursukov, N., Kim, I., & Viksnin, I. (2022). "Study on Destructive Informational Impact in Unmanned Aerial Vehicles Intergroup Communication" in Symmetry. 2022. Vol. 14, no. 8, pp. 1-18. URL. **Q2**. *Published in Special Issue*

[J8] Viksnin, I., Marinenkov, E., & **Chuprov, S.** (2022). "A Game Theory approach for communication security and safety assurance in cyber-physical systems with Reputation and Trust-based mechanisms" in Scientific and Technical Journal of Information Technologies, Mechanics and Optics, vol. 22, no. 1, pp. 47–59. doi: 10.17586/2226-1494-2022-22-1-47-59. URL

[J9] Khokhlov, I., Reznik, L., & **Chuprov, S.** (2020). "Framework for Integral Data Quality and Security Evaluation in Smartphones" in IEEE Systems Journal, vol. 15, no. 2, pp. 2058-2065, doi: 10.1109/JSYST.2020.2985343. URL. **Q1**

[J10] **Chuprov, S.**, Viksnin, I., Kim, I., Tursukov, N., & Nedosekin, G. (2020). Empirical Study on Discrete Modeling of Urban Intersection Management System. International Journal of Embedded and Real-Time Communication Systems (IJERTCS), vol. 11(2), pp. 16-38, doi: 10.4018/IJERTCS.2020040102. URL

[J11] **Chuprov, S.**, Viksnin, I., Kim, I., Marinenkov, E., Usova, M., Lazarev, E., Melnikov, T., & Zakoldaev, D. (2019). Reputation and Trust Approach for Security and Safety Assurance in Intersection Management System. Energies, 12(23), 4527, doi: 10.3390/en12234527. URL. **Q1**

REFEREED CONFERENCES' PROCEEDINGS AND PRESENTATIONS

[C1] Peña, A. & **Chuprov, S.** (2026). "Improving PID-Based Aggregation in Federated Learning for Consumer Applications" in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366291. URL

[C2] Zatsarenko, R., **Chuprov, S.**, Korobeinikov, D., Reznik, L., & Peña, A. (2026). "How to Improve Federated Learning in Consumer Applications? Detect and Exclude Bad Clients" in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366313. URL

[C3] Peredo, A. & **Chuprov, S.** (2026). "Improving Generalization with Harmonic Aggregation in Personalized Federated Learning" in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366390. URL

[C4] Zatsarenko, R., **Chuprov, S.**, Korobeinikov, D., Reznik, L., & Peña, A. (2025). "MADE-PI: Framework for Effective Anomaly Detection in Federated Learning Applications" in 2025 International Conference on Machine Learning and Applications (ICMLA). 2025, pp. 446-451, doi: 10.1109/ICMLA66185.2025.00067. URL

[C5] Fonseca, P., **Chuprov, S.**, Solis, J. P., & Marquez, G. (2025). "Improving Lean Construction by Integrating ML for Contractor Performance Prediction" in 2025 38th Conference of Open Innovations Association (FRUCT). 2025. doi: 10.23919/FRUCT67853.2025.11239221. URL

[C6] Chan, E. E. N. & **Chuprov, S.** (2025). "Bayesian Optimization of PID Coefficients for Robust Federated Learning" in TACCSTER 2025 Proceedings. 2025. doi: 10.26153/tsw/62178. URL

[C7] Hong, L., **Chuprov, S.**, & Wan, Y. (2025) "Mitigating Health Misinformation Through Human-AI Collaboration" in AMCIS 2025 Proceedings. URL

[C8] **Chuprov, S.**, Korobeinikov, D., Zatsarenko, R., & Reznik, L. (2025) "No trust, no learning? Improving Federated Learning Security and Robustness with Reputation and Trust" in Data Science Workshop "From Theory to Practice". URL

[C9] Korobeinikov, D., **Chuprov, S.**, & Reznik, L., (2024). "Towards More Robust Federated Learning with Medical Imaging Model Anomaly Detection" in 2024 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2024, pp. 1-4. URL

[C10] Narayanan, A., **Chuprov, S.**, Reznik, L., Zatsarenko, R., & Korobeinikov, D. "Intelligent Soccer Event Detection and Highlights Generation with Broadcast Cues Integration" in 23rd International Conference on Machine Learning and Applications (ICMLA'24), Miami, FL, USA, 2024, pp. 554-559, doi: 10.1109/ICMLA61862.2024.00081. URL

[C11] Patel, H., **Chuprov, S.**, Korobeinikov, D., Zatsarenko, R., & Reznik, L. (2024). "Improving Federated Learning Security with Trust Evaluation to Detect Adversarial Attacks" in 19th Annual Symposium on Information Assurance (ASIA' 24), 2024, pp. 37-43. URL

[C12] Korobeinikov, D., **Chuprov, S.**, Zatsarenko, R., & Reznik, L., (2024). "Federated Learning Robustness on Real World Data in Intelligent Transportation Systems" in 19th Annual Symposium on Information Assurance (ASIA' 24), 2024, pp. 30-36. URL

[C13] **Chuprov, S.**, Zatsarenko, R., Korobeinikov, D., & Reznik, L. (2024). "Robust Training on the Edge: Federated vs. Transfer Learning for Computer Vision in Intelligent Transportation Systems" in 2024 IEEE World AI IoT Congress (AIIoT), Seattle, WA, USA, 2024, pp. 172-178, doi: 10.1109/AIIoT61789.2024.10578970. URL

[C14] Zatsarenko, R., **Chuprov, S.**, Korobeinikov, D., & Reznik, L. (2024). "Trust-Based Anomaly Detection in Federated Edge Learning" in 2024 IEEE World AI IoT Congress (AIIoT), Seattle, WA, USA, 2024, pp. 273-279, doi: 10.1109/AIIoT61789.2024.10578967. URL

[C15] Kovtun R., **Chuprov, S.**, Gataullin, R., Ruchkan, A., Alhasan, A., & Viksnin, I. (2024). "A Modern Approach to High Dynamic Range Image Processing with Machine Learning Architectures" in 2024 XXVII International Conference on Soft Computing and Measurements (SCM), 2024, pp. 207-212, doi: 10.1109/SCM62608.2024.10554097. URL

[C16] Garifullin M., Turushev, T., Tursukov, N., & **Chuprov, S.** (2024). "Extended Formulation of the Problem of Terrain Exploration using a Multi-agent System" in 2024 XXVII International Conference on Soft Computing and Measurements (SCM), 2024, pp. 207-212, doi: 10.1109/SCM62608.2024.105
54236. URL

[C17] **Chuprov, S.**, Mahajan, S., Zatsarenko, R., Reznik, L. & Ruchkan, A. (2023). "Are Industrial ML Image Classifiers Robust to Withstand Adversarial Attacks on Videos?" in 2023 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2023, pp. 1-4, doi: 10.1109/WNYISPW60588.2023.10349595. URL

[C18] Zatsarenko, R., **Chuprov, S.**, Marathe, C.A., Hyland, M., & Reznik, L. (2023). "Are Industrial ML Image Classifiers Robust to Data Affected by Network QoS Degradation?" in 2023 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2023, pp. 1-4, doi: 10.1109/WNYISPW60588.2023.10349560. URL

[C19] Neverov, E., Viksnin, I., & **Chuprov, S.** (2023) "The Research of AutoML Methods in the Task of Wave Data Classification" in 2023 XXVI International Conference on Soft Computing and Measurements (SCM), 2023, pp. 156-158, doi: 10.1109/SCM58628.2023.10159058. URL

[C20] Tursukov N., Viksnin I., Neverov E., Sheinman E., & **Chuprov S.** (2023) "Evaluation of the Effectiveness of Neural Networks Based on the Criteria for Completing the Object Classification Task" in 2023 XXVI International Conference on Soft Computing and Measurements (SCM), 2023, pp. 120-122, doi: 10.1109/SCM58628.2023.10159070. URL

[C21] **Chuprov, S.**, Bhatt, K.M., & Reznik, L. (2023). "Federated Learning for Robust Computer Vision in Intelligent Transportation Systems" in 2023 IEEE Conference on Artificial Intelligence (CAI), Santa Clara, CA, USA, 2023, pp. 26-27, doi: 10.1109/CAI54212.2023.00019. URL

[C22] **Chuprov, S.**, Memon, M., & Reznik, L. (2023). "Federated Learning with Trust Evaluation for Industrial Applications" in 2023 IEEE Conference on Artificial Intelligence (CAI), Santa Clara, CA, USA, 2023, pp. 347-348, doi: 10.1109/CAI54212.2023.00153. URL

[C23] **Chuprov, S.**, Reznik, L., & Grigoryan, G. (2022). "Study on Network Importance for ML End Application Robustness" ICC 2023 - IEEE International Conference on Communications, Rome, Italy, 2023, pp. 6627-6632, doi: 10.1109/ICC45041.2023.10279698. URL

[C24] **Chuprov, S.**, Satam, A. N., & Reznik, L. (2022). "Are ML Image Classifiers Robust to Medical Image Quality Degradation?" in 2022 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2022, pp. 1-4, doi: 10.1109/WNYISPW57858.2022.9983488. URL

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